# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science & Engineering

## ASSESSMENT TOOL 3 - VIVA-VOCE QUESTIONS

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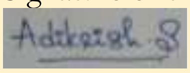
|---|---|---|---|
| **Course Code** | **DSA0404** | **Course Title** | **Fundamentals of Data Science** |
| **CO Assessed** | **CO4 – Viva-Voce** | **Assessment** | **Assessment Tool 3: Viva-Voce** |
| **Weightage** | **20% of CIA** | **Total Marks** | **100** |
| **CO4 Statement** | **To apply the statistical inferences such as testing of hypothesis and point estimates for the given data** | | |
| **Student Name** | **Aditya Krishnan S** | **Reg. No.** | **192424069** |
| **Section/Batch** | **2024** | **Date** | **21 August 2026** |

**Code of Conduct:**

I, ADITYA KRISHNAN S/192424069, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) | Marks |
|---|---|---|---|---|---|---|
| Understanding of Question | 25 | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly | |
| Application of Concepts | 25 | Effectively applies relevant concepts to analyse the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts | |
| Analysis | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning | |
| Solution Design | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided | |
| Clarity | 10 | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand | |

## A. Introduction to Statistical Inference

**1. What is statistical inference?**
Statistical inference is the process of using sample data to draw conclusions about a population. It includes estimation and hypothesis testing.

**2. Why is statistical inference important in data science?**
It helps data scientists estimate unknown population characteristics, test assumptions, and make decisions from sample data.

**3. What is the difference between descriptive statistics and inferential statistics?**

→ Descriptive statistics: Summarizes and describes observed data.
→ Inferential statistics: Uses sample data to make conclusions about a larger population.

**4. What is a population in statistics?**
A population is the complete set of individuals, objects, or observations of interest in a study.

**5. What is a sample?**
A sample is a subset of the population selected for analysis.

**6. Why do data scientists usually work with samples instead of complete populations?**
Studying an entire population may be time-consuming, costly, or impractical. A representative sample provides useful information more efficiently.

**7. What is a population parameter?**
A population parameter is a numerical characteristic of a population, such as population mean (\mu) or proportion (p).

**8. What is a sample statistic?**
A sample statistic is a numerical measure calculated from sample data, such as sample mean (\bar{x}) or sample proportion (\hat p).

**9. Give a real-world example where statistical inference is used in industry.**
A company may survey a sample of customers to estimate the satisfaction level of all its customers.

**10. What are the two major activities involved in statistical inference?**
The two major activities are:

→ Estimation
→ Hypothesis testing

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## B. Frequentist Approach

**11. What is the frequentist approach to statistical inference?**
It is an approach where probability is interpreted through long-run frequencies, and unknown parameters are treated as fixed quantities.

**12. What is the basic idea behind frequentist statistics?**
The basic idea is to use repeated-sampling behavior of statistics to estimate parameters and test hypotheses.

**13. How does the frequentist approach differ from the Bayesian approach?**
The frequentist approach treats parameters as fixed but unknown, while the Bayesian approach treats parameters as uncertain and assigns probability distributions to them.

**14. In the frequentist approach, what is considered fixed: the parameter or the data?**
The population parameter is fixed, while the observed data and sample statistics are considered variable across samples.

**15. What does long-run frequency mean in frequentist statistics?**
It is the proportion of times an event occurs over a very large number of repeated trials under the same conditions.

**16. Why is random sampling important in the frequentist approach?**
Random sampling helps ensure that the sample is representative of the population and supports valid statistical inference.

**17. Can a population parameter be considered a random variable in the frequentist approach? Why?**
No. It is considered a fixed but unknown value; randomness comes from the sampling process.

**18. Give an example of a frequentist problem in a data science application.**
Testing whether the average response time of a software application differs from a specified target is a frequentist problem.

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## C. Point Estimation

**19. What is a point estimate?**
A point estimate is a single numerical value used to estimate an unknown population parameter.

**20. What is an estimator?**
An estimator is a rule or formula calculated from sample data that is used to estimate a population parameter.

**21. What is the difference between an estimator and an estimate?**
An estimator is the method or formula; an estimate is the numerical value obtained after applying it to a sample.

**22. What is the point estimator for the population mean?**
The sample mean is the point estimator:

$$\bar{x} = \frac{\sum x_i}{n}$$

**23. How do you estimate a population proportion from a sample?**
Use the sample proportion:

$$\hat{p} = \frac{x}{n}$$

where (x) is the number of successes.

**24. What is the point estimate of population variance?**
The sample variance is used:

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

**25. Why is the sample mean commonly used to estimate the population mean?**
Because the sample mean is generally an unbiased estimator of the population mean and becomes more reliable as sample size increases.

**26. What properties should a good estimator have?**
A good estimator should ideally be unbiased, consistent, and efficient.

**27. What is an unbiased estimator?**
$E(\hat{\theta}) = \theta$

**28. What is meant by the bias of an estimator?**
Bias is the difference between the expected value of an estimator and the true parameter.
$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$

**29. What is the difference between bias and variance of an estimator?**
Bias: Systematic difference from the true parameter.
Variance: Variation of the estimator across different samples.

**30. Why can two different samples produce different point estimates for the same population parameter?**
Because of sampling variability. Different random samples contain different observations and therefore produce different statistics.

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## D. Variability in Estimates

**31. Why does a point estimate have uncertainty?**
A point estimate is calculated from a sample rather than the entire population, so it varies from sample to sample.

**32. What is the sampling distribution of an estimator?**
It is the probability distribution of an estimator obtained from all possible samples of a given size.

**33. What is the standard error?**
Standard error measures the standard deviation of the sampling distribution of an estimator.
For a sample mean:
$$SE = \frac{\sigma}{\sqrt{n}}$$

**34. How is standard error different from standard deviation?**
Standard deviation measures variability among individual observations, whereas standard error measures variability of a sample statistic across samples.

**35. How does increasing the sample size affect the standard error of the sample mean?**
Increasing sample size reduces the standard error:
$$SE = \frac{\sigma}{\sqrt{n}}$$

**36. Why does variability in an estimate decrease when the sample size increases?**
A larger sample contains more information about the population, making the estimate more stable and less affected by sampling variation.

**37. What is the relationship between standard error and confidence intervals?**
A larger standard error produces a wider confidence interval, while a smaller standard error produces a narrower interval.

**38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.**
No. Different sample means can occur naturally because of sampling variability. Both samples may be valid.

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## E. Confidence Intervals

**39. What is a confidence interval?**
A confidence interval is a range of plausible values for an unknown population parameter, calculated from sample data.

**40. Why is a confidence interval more informative than a point estimate?**
A point estimate gives only one value, while a confidence interval also shows the uncertainty or precision of the estimate.

**41. What does a 95% confidence level mean in the frequentist approach?**
If the sampling procedure is repeated many times, approximately 95% of the resulting confidence intervals would contain the true population parameter.

**42. What is the difference between a confidence level and a confidence interval?**
Confidence level: The chosen degree of confidence, such as 95%.
Confidence interval: The actual numerical range calculated from the sample.

**43. What factors determine the width of a confidence interval?**
The width depends mainly on the confidence level, sample size, and variability in the data.

**44. How does increasing the sample size affect the width of a confidence interval?**
Increasing sample size generally reduces the standard error and makes the confidence interval narrower.

**45. How does increasing the confidence level from 95% to 99% affect the confidence interval?**
The interval becomes wider, because greater confidence requires a larger margin of error.

**46. What is the difference between a z-confidence interval and a t-confidence interval?**
A z-interval uses the standard normal distribution, while a t-interval uses the t-distribution, particularly when population standard deviation is unknown.

**47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?**
Use the t-distribution when the population standard deviation is unknown, especially for smaller samples.

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## F. Hypothesis Testing Using Confidence Intervals

**48. What is hypothesis testing?**
Hypothesis testing is a statistical method used to evaluate a claim about a population parameter using sample evidence.

**49. What are the null hypothesis ($H_0$) and alternative hypothesis ($H_1$)?**
($H_0$): The initial claim or assumption about the population parameter.
($H_0$): The competing claim that provides evidence against ($H_0$).

**50. How can a confidence interval be used to make a decision about a hypothesis?**
If the hypothesized parameter lies inside the confidence interval, we generally fail to reject ($H_0$). If it lies outside, we reject ($H_0$) at the corresponding significance level.

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